2D Projectile Motion: Launch At Arbitrary Angles

Name: _____ Date: _____

SYNOPSIS

A ball launched at an angle is really two separate motions running on one clock: a horizontal motion at constant velocity, and a vertical motion in freefall. They don't interact — gravity never touches the horizontal component, and the horizontal component has no say in how fast the ball falls. So the first move is always the same: **resolve the launch velocity**. Splitting $\mathbf{V_0}$ into $V_{0x} = V_0\cos\theta$ and $V_{0y} = V_0\sin\theta$ lets us use the freefall equations from Note 4 on the vertical piece and the constant-velocity equation on the horizontal piece.

Everything in this note is a **level-ground launch**: the ball lands at the same height it was launched from. That makes the flight symmetric. It rises for exactly half the flight, falls for the other half, and lands with the same speed it left with. We find the **flight time first**, because it depends only on V_{0y}, and then the maximum height **H** and the range **D** follow from it. The numbers we carry the whole way through are the launcher from class: 5 m/s at 30°.

PERFORMANCE EXPECTATIONS

By the end of this note, you will be able to...

- PE1 **Resolve a launch velocity into components** — $V_{0x}=V_0\cos\theta$ and $V_{0y}=V_0\sin\theta$ — and explain why the two motions can be analysed separately.
- PE2 **Write the x and y equations** for an angled launch: $d_x=V_{0x}t$, and $d_y=V_{0y}t-\frac{1}{2}gt^2$ with $V_y=V_{0y}-gt$.
- PE3 **Find the flight time** of a level-ground launch, $t_{flight}=2V_{0y}/g$, and explain why it depends only on V_{0y}.
- PE4 **Find the maximum height H** from the time to the apex, where $V_y=0$ and $t_{apex}=t_{flight}/2$.

PE5 **Find the range D=$V_{0x}t_{flight}$**, and describe how the ball lands — same speed, angled θ below horizontal.

You already know this

Nothing here is brand new — you're combining three things you've already built:

📐 **Resolving a vector (N2 / N3)**

Adjacent side → cosine, opposite side → sine. V_0 splits the same way any vector does.

🍎 **Freefall (N4)**

The vertical piece obeys the same $v=v_o+at$ and $d=d_o+v_ot+½at^2$, with $a=-g=-9.8$ m/s² always.

🚗 **Constant velocity (M1)**

Nothing pushes sideways, so the horizontal piece never changes: $d_x=V_{0x}t$ for the whole flight.

In this note, we work through

I. How do we split a launch velocity into horizontal and vertical components?

II. Why does the flight time depend only on the vertical component?

III. How long does the ball take to reach the top, and how high does it go?

IV. How far does it land from where it started, and how is it moving when it lands?

V. Your turn: a kicked soccer ball, start to finish.

Predict first: on level ground, two balls leave at the same speed V_0 — one at 30° above horizontal, the other at 60°. Which one stays in the air longer?

YOUR PREDICTION, AND WHY

Resolve the launch velocity

A ball leaves level ground with speed V_0 at angle θ above horizontal. The horizontal side of the triangle is *adjacent* to θ, so it takes the cosine; the vertical side is *opposite* θ, so it takes the sine:

$$V_{0x} = V_0\cos\theta \qquad V_{0y} = V_0\sin\theta$$

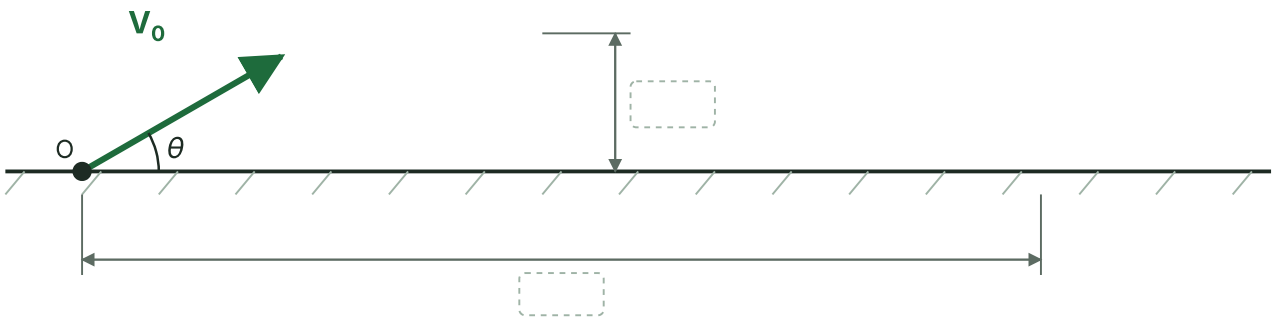


Figure 1. Draw V_{0x} and V_{0y} on the launch triangle and label each with cosθ or sinθ. Sketch the path back to the ground, mark the apex and write what V_x and V_y are there, fill in H and D, and draw the landing velocity.

The launcher from class: V_0=5 m/s at θ=30°.

Now that V_0 is resolved into a horizontal and a vertical piece, and the two pieces are independent of each other, we can analyse the x motion and the y motion separately.

Two motions, one clock

Measure everything from the launch point, so d_0=0 in both directions. The acceleration is zero sideways and a_y=−g=−9.8 m/s² vertically — always, on the way up, at the top, and on the way down.

	Horizontal (x)	Vertical (y)
Position	$d_x = V_{0x}t = (V_0\cos\theta)t$	$d_y = V_{0y}t - \frac{1}{2}gt^2 = (V_0\sin\theta)t - 4.9t^2$
Velocity	$V_x = V_{0x} = V_0\cos\theta$ (never changes)	$V_y = V_{0y} - gt = V_0\sin\theta - 9.8t$
5 m/s at 30°		

The same t appears in both columns — that is the clock the two motions share. And because d_y has a t^2 term while d_x is just proportional to t, the path is a **parabola**.

First, find the flight time

On level ground the trip down mirrors the trip up, so the ball comes back to the ground moving downward at the speed it left with: $V_y = -V_{0y}$ at landing. Put that into the vertical velocity equation:

The launcher from class:

Notice: V_{0x} never appears. The flight time is set by the vertical motion alone, so a steeper launch at the same speed (bigger V_{0y}) stays up longer. The horizontal speed only decides how far sideways the ball gets in that time. Same answer from the position equation: the ball is back on the ground when $d_y = 0$, so $0 = 2.5t - 4.9t^2$, which gives $t = 0$ (the launch) or $t = 2.5/4.9 \approx 0.51$ s (the landing).

Time to the apex

At the top of the arc the ball is moving purely sideways for an instant, so $V_y = 0$ there:

That is the symmetry of a level-ground flight: half the time going up, half coming down.

Maximum height, H

H is the vertical position at t_{apex}. Put t_{apex} into the vertical position equation:

Shortcut, same answer: putting $t_{apex}=V_{0y}/g$ into that equation and simplifying gives $H=V_{0y}^2/(2g)$. Check: $2.5^2/(2\cdot 9.8)=6.25/19.6\approx 0.32$ m. Use whichever you trust; both come from the same physics.

Range, D

The horizontal motion never changes, so the ball travels sideways at V_{0x} for the whole flight time:

In class we measured the launcher's landing spot at about 2.25 m — within a few centimetres of the prediction.

How it lands

At t_{flight}, find V_x and V_y. Compare them with the launch components. How does the ball land?

Key equations — level-ground launch

	Horizontal component	Vertical component
Velocity (m/s)	$V_x = V_0\cos\theta$	$V_{0y} = V_0\sin\theta$
Flight time, t_{flight} (s)	—	$t_{flight} = 2V_{0y}/g = 2V_0\sin\theta/g$
Max. displacement (m)	$D = d_{x,max} = V_{0x}t_{flight} = (V_0\cos\theta)t_{flight}$	$H = d_{y,max} = V_{0y}(t_{flight}/2) - \frac{1}{2}g(t_{flight}/2)^2$

Try it in the launcher: at the same speed, 30° and 60° land in the same spot. The 60° ball goes higher and stays up longer, but it moves sideways more slowly, and the two effects cancel exactly. Any pair of angles that add to 90° shares a range on level ground; 45° gives the longest range of all.

YOUR TURN — A KICKED SOCCER BALL

A soccer ball is kicked from level ground at V_0=12 m/s, 40° above horizontal. Use g=9.8 m/s². Work each part in your notes, then open the working to check.

1. Resolve V_0 into V_{0x} and V_{0y}.
2. Find the flight time and the time to the apex.
3. Find the maximum height H.
4. Find the range D.

YOUR WORKING

Where this fits: every result in this note assumes the ball lands at the same height it left from. In Note 4 the launch was horizontal from above the ground, so the drop height, not V_{0y}, set the time. Here the launch and the landing are level, so the time comes from V_{0y} alone — find it first, and everything else follows.

🎟 Exit ticket

Four quick questions — answer on this sheet.

Q1. Which velocity component determines how long a projectile stays in the air?

- ○ Only the horizontal component, V_{0x}.
- ○ Only the vertical component, V_{0y}.
- ○ Both components equally.
- ○ Neither — flight time is fixed by g alone.

Q2. At the apex of the trajectory…

- ○ Both V_x and V_y are zero.
- ○ $V_y = 0$, and V_x still equals $V_0 \cos\theta$.
- ○ $V_x = 0$, and V_y is at its maximum.
- ○ The acceleration is zero for an instant.

Q3. **A ball is launched from level ground at V_0=10 m/s, 30° above horizontal (g=9.8 m/s²). What is its flight time?**

_____ s

Q4. **On level ground, two balls leave at the same speed, one at 30° and one at 60°. Which one lands farther away?**

- The 30° ball.
- The 60° ball.
- They land the same distance away.
- It depends on the launch speed.

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